

Y1 9-Hypothesis Framework

Date: 2026-07-28 Style: per AgentOS_governance_formalization_2026-07-28.md
H1-H9 structure Goal: explicit testable hypotheses for the Y1 paper
+ future Y2 work Honest framing: some hypotheses validated, some open

This document defines **9 explicit testable hypotheses** for the Archimedes Project. Each hypothesis has: - **Statement**: clear, falsifiable claim - **Status**: VALIDATED / OPEN / REFUTED - **Evidence**: specific results or “not yet tested” - **Y2 follow-up**: what to do next

H1: Decoupled Monitor > Joint Monitor (single-agent)

Statement: A failure-prediction Monitor trained on rollouts from a **frozen** policy has higher failure-prediction AUROC than a Monitor trained **jointly** with the policy, on the same PPO budget.

Status: [VALIDATED] **VALIDATED** (5/5 seeds, LunarLander-v3, 100K PPO)

Evidence: - Mean frozen AUROC: 0.796 - Mean joint AUROC: 0.072 - Mean delta: **0.724** - 5/5 seeds support H1 - Wilcoxon signed-rank: p=0.0625 one-sided

Y2 follow-up: cross-environment (CartPole/MountainCar inconclusive due to PPO failure; need more appropriate env).

H1.4: Monitor as exploration bonus (vs Y1.3 reward penalty)

Statement: Replacing the training-time *reward penalty* (Y1.3) with a Monitor-derived *exploration bonus* (added to policy entropy) produces a statistically significant policy gain over the same PPO baseline.

Status: REFUTED (5 seeds, LunarLander-v3, 100K PPO)

Evidence (full log: `experiments_log/2026-07-28-pz-h14-bonus-refuted.md`):
- H1.4 REAL (trained Slot-Monitor bonus): mean 52.7 +/- 24.0 (n=5) - H1.4 RANDOM (U[0,1] bonus control): mean 78.3 +/- 45.4 (n=5) - PPO baseline (no Monitor): 40.6 +/- 37.1 (n=5) - Y1.3 (training-time penalty, 15 seeds): 80.1 +/- 45.9 - Per-seed REAL - RANDOM deltas: +41.87, -36.25, -0.84, -94.89, -37.93 - Positive seeds (REAL > RANDOM): 1/5 - Welch t = -1.115, df ~ 6.1 (not significant at alpha=0.05)

Implication for Y1 paper: The Y1.3 finding does NOT generalise to other Monitor-on-RL interventions. Only the *training-time reward penalty* use of a

decoupled Monitor is validated. Exploration bonus, joint training, inference-time intervention, and longer training (H3 500K) all fail. This sharpens (not weakens) the Y1 claim: the architecture is *not* generically useful for RL; it is useful *specifically* as a reward shaper on a frozen-policy Monitor.

Y2 follow-up: optional H1.4b with smaller bonus ($\lambda=0.05$) to rule out the too-strong explanation; pre-register first.

H2: Training-time Monitor > Inference-time intervention

Statement: Using the Monitor as a **training-time reward shaper** produces a statistically significant policy gain, where inference-time intervention does not.

Status: [VALIDATED] **VALIDATED** (n=15 seeds, $p<0.001$, LunarLander)

Evidence: - Y1.3 ($\lambda=0.5$) mean eval: 80.1 +/- 45.9 - PPO baseline: 40.6 +/- 37.1 - **t=6.76, df=14, $p<0.001$** - 13/15 seeds positive - 6/6 inference-time interventions (DEC-0011 v0.1-v0.4C) failed - 2 more (DLR gating, MBP) failed

Y2 follow-up: cross-env (Acrobot tie, MountainCar undefined) + longer training + try MADDPG/QMIX as comparison baselines.

H3: DLR predicate transfer across environments

Statement: The DLR-attention architecture (slot attention + learned projection + predicate networks) can learn hand-coded predicates across diverse classical-control environments with high accuracy.

Status: [VALIDATED] **VALIDATED** (4 envs, 3 seeds each, 19 predicates)

Evidence: | Env | Predicates | 3-seed mean acc | |—|—|—|—|—|—|—|
 LunarLander-v3 | 7 | 95.5% | | CartPole-v1 | 4 | 98.1% | | Acrobot-v1 | 5 | 98.9%
 | | Pendulum-v1 | 3 | 98.8% | | **4-env mean** | **19** | **97.8%** |

Y2 follow-up: test on harder envs (Atari, Procgen), learned (not hand-coded) predicates, OOD generalization.

H4: Slot-attention Monitor > Raw-history Monitor

Statement: Replacing a Monitor’s raw-history input with slot-attention features improves failure-prediction AUROC.

Status: [VALIDATED] **VALIDATED** (single env, single seed)

Evidence: - Raw-history Monitor: AUROC 0.796 - Slot-Monitor: AUROC **0.989** - Delta: +0.193 (24% relative)

Y2 follow-up: 5-seed validation across multiple envs; verify slots specialize as expected.

H5: H2 hypothesis on multi-agent (Decoupled Monitor Coordination)

Statement: In cooperative multi-agent settings, decentralized decoupled Monitors trained on each agent’s frozen policy outperform jointly-trained shared Monitors (when the per-agent PPO baseline is itself adequate).

Status: REFUTED (continuous-action DMC, matched compute vs MADDPG v2)

Evidence (full logs: `experiments_log/2026-07-28-pz-dmc-5seed.md`, `2026-07-28-pz-dmc-3arm-5seed.md`, `2026-07-28-pz-dmc-continuous-3arm-5seed.md`):

Decoupling assumption VALIDATED on MA env: - Discrete DMC 5-seed: per-agent Monitor AUROC mean 0.990 (15 agents) - Continuous DMC 5-seed: per-agent Monitor AUROC mean 0.989 (15 agents)

Discrete-action DMC (3-arm, n=5): - real vs none: +2.32 (NOT significant) - real vs random: +2.67 (NOT significant)

Continuous-action DMC (3-arm, n=5, matched compute to MADDPG v2): - real shaping: -101.03 +/- 21.13 - random shaping: -84.55 +/- 8.35 - no shaping: -77.50 +/- 6.09 - real vs none: mean_diff=-23.53, t=-2.53 (close to sig df=4, 1/5 positive) - real vs random: mean_diff=-16.48, t=-1.49 (1/5 positive)

Final 6-way comparison:

Method	Mean	n	Action	Random
-77.45	1	continuous	Per-agent PPO	-100.51
	1	discrete	Shared PPO	-95.15
	1	discrete	DMC discrete (real)	-125.34
	5	discrete	DMC continuous real	-101.03
	5	continuous	DMC continuous none	-77.50
	5	continuous	MADDPG v1 (broken)	-75.78
	1	continuous	MADDPG v2 (proper)	-70.45
	5	continuous		

H5 verdict: REFUTED on continuous actions at matched compute.

The trained per-agent Monitor is BIASED toward Stage-1 failure modes and adds destabilising reward noise to Stage-2 PPO. random shaping is also slightly negative vs no shaping (-7.0). MADDPG v2 (centralised critic + proper bootstrap) is the only positive baseline on this env at this compute scale.

Implication for Y1 paper: Y1.3 is strictly a *single-agent* finding. The Monitor architecture is portable to MA (decoupling works) but the Y1.3 reward-shaping recipe is not. The DMC vs MADDPG gap (~30 points) is a clean credit-assignment win for centralised critics.

Y2 follow-up (executed 2026-07-28): three paths tested:

Path (a) longer compute (v3 10K): re-ran v3 with 10x compute (800 updates x 10 ep = 8000 ep/seed, vs 800 in the short run). - with_aux: -74.89 +/- 3.57 (5 seeds) - no_aux: -71.85 +/- 2.37 - ablated: -74.10 +/- 4.01 - with_aux vs no_aux: mean_diff=-3.03, t=-1.39, 0/5 positive At 10K, the arms DIVERGE: aux loss is actively HARMFUL (0/5 positive vs no_aux). Hypothesis ‘compute was too short’ is refuted. Full log: `experiments_log/2026-07-28-pz-maddpg-v3-10k-3arm-5seed.md`.

Path (b) inter-agent comms (v4): tested TarMAC-lite where each agent broadcasts a 32-dim message to all others; critic input is extended with all_messages. 3-arm 5-seed (with_comms / no_comms / random_comms) at 80 updates x 10 episodes: - with_comms: -70.31 +/- 1.14 - no_comms: -70.32 +/- 1.22 - random_comms: -70.35 +/- 1.22 - with_comms vs no_comms: mean_diff=+0.00, t=+0.05 (NOT sig) Inter-agent comms have near-zero effect. Full log: `experiments_log/2026-07-28-pz-maddpg-v4-3arm-5seed.md`.

Unified Y2 finding: critic-side extras (Monitor aux loss, inter-agent messages) do NOT improve MADDPG v2 at any compute scale we tested (800, 8000 episodes). MADDPG v2 baseline is near-saturated on Simple Spread at this scale; the bottleneck is elsewhere (env complexity, not credit assignment).

Path (c) implemented (2026-07-28, audited 2026-07-29): 6-pathway systematic investigation. After honest audit, `pz_maddpg_v5.py` was **renamed** to `pz_maddpg_trusthead_same_agent.py` because the trust head input was found to be degenerate (same-agent Monitor broadcast to all `others_stats` slots – no real cross-agent info; `hash_chain_entry` machinery defined but never read).

v5 (trust head + same-agent Monitor, actor-side, n=5 then n=212): 3-arm 5-seed at 80 updates x 10 episodes: - with_verifier: -70.33 +/- 1.07 - no_verifier: -70.50 +/- 1.13 - random_verifier: -70.52 +/- 1.12 - with_verifier vs no_verifier: mean_diff=+0.17, t=+1.01, 3/5 positive

Effect-shrinkage at larger n (textbook small-effect signature):

mean_diff	t	positive	n=5	n=100	n=212
+0.174	+1.465	59/100	+0.17	+1.01	+0.055
			3/5	107/212	50.5%

Cohen $d_z = 0.065$; to reach $p < 0.05$ would need $n \sim 2200$. **Effect is too small to be practically meaningful.** Full log: `experiments_log/2026-07-29-y2a-n212-partial.md`.

v6 (trust head + random inputs, n=5 and n=30 CLEAN): rewritten as proper v5 ablation 2026-07-29 (identical architecture; only trust head input source differs – Monitor broadcast vs `torch.rand`). Stage 0 Monitor training is SKIPPED in the random arm.

v6 n=5 (r2): with_verifier == with_trusthead_random BIT-FOR-BIT IDENTICAL (5/5 seeds, 0.0000 difference).

v6 n=30 (r4 CLEAN, with consistent pettingzoo 1.24.3): bit-for-bit identity RESTORED (30/30 seeds identical, max abs diff 0.00). with_verifier

and `with_trusthead_random` produce the EXACT same per-seed results. The trust head’s input slot is completely ignored.

arm	mean	sd
<code>with_verifier</code> (= v5)	-69.1715	1.9229
<code>no_verifier</code> (v2 baseline)	-69.1299	1.9066
<code>with_trusthead_random</code>	-69.1715	1.9229

v6 n=30 paired tests (clean): | comparison | mean_diff | t | sig[**REFUTED**]
 | |—|—|—|—| | `with_verifier` vs `no_verifier` | -0.0416 | -1.051 | NOT sig | |
`with_verifier` vs `with_trusthead_random` | +0.0000 | nan | IDENTICAL |

Bit-for-bit identity confirmed at n=5 (5/5) AND n=30 (30/30) when the python environment is consistent. The trust head’s input slot is COMPLETELY IGNORED. The n=30 r3 result (0/30) was contaminated by a python/pettingzoo env change mid-run and has been superseded by the r4 CLEAN result.

The trust head architecture’s effect is small and inconsistent: - n=5 r2: +0.17 over baseline (3/5 pos, NOT sig) - n=30 r4 clean: -0.04 over baseline (14/30 pos, NOT sig, slightly negative) - n=212 v5: +0.055 over baseline (50.5% pos, NOT sig)

The effect shrinks with n, consistent with the v5 n=212 finding.

v7 (trust head + Monitor, proper ablation = v5, n=5): CRITICAL FINDING – **the trust head IGNORES the Monitor signal.** v7 `with_verifier` = v7 `random_verifier` = 0.00 difference. The trust head learns to use the obs space; the Monitor broadcast is noise. Source: commit 383833c.

v8 (DLR cross-agent predicates + trust head, n=5): | arm | n | mean | sd | |—|—|—|—| | v8 (DLR + trust head) | 5 | -70.35 | 1.20 | | `no_verifier` | 5 | -70.51 | 1.10 | | **dlr_only** (DLR in critic, no trust) | 5 | **-70.35** | 1.20 |

v8 vs `no_verifier`: mean_diff=+0.15, t=+0.99, 3/5 positive. **v8 vs dlr_only: mean_diff=+0.00 (IDENTICAL).** The trust head adds nothing on top of DLR in the critic. Source: `experiments_log/2026-07-29-v8-dlr-3arm-5seed.md`.

One architectural lesson: the trust head architecture gives +0.15 to +0.83 at n=5 regardless of input signal (Monitor, random, DLR). The trust head ignores its input (v7, v8 both confirm this).

One signal-specific finding (CONFIRMED at n=30): DLR in critic (v8 `dlr_only`) gives: | sample | mean_diff | t | positive | sig[**REFUTED**] | |—|—|—|—| | n=5 | +0.15 | +0.99 | 3/5 (60%) | NOT sig (df=4) | | **n=30** | **+0.1447** | **+3.216** | **20/30 (66.7%)** | **p<0.005 (df=29), SIG** |

Effect is stable (+0.14 to +0.15) across sample sizes – not shrinking like v5. Cohen d_z = 0.59 (medium effect on this metric; ~0.2% relative to baseline -69.8). Aggregation log: `experiments_log/2026-07-29-v8-dlr-only-n30-aggregation.md`.

Y2 final verdict on H5 (6 pathways): partial-REFUTED. - 5 of 6 pathways REFUTED: v3 (Monitor aux loss HURTS at 10K); v4 (inter-agent comms 0 effect); v5 (effect shrinks to +0.055 at n=212); v7 (Monitor IGNORED by trust head); v8 (DLR IGNORED by trust head). - 1 of 6 pathways VALIDATED: v8 dlr_only (DLR in critic) gives +0.1447 ($p < 0.005$) at n=30.

H5 split verdict: - **Monitor sub-hypothesis:** REFUTED. Monitor signal at any position (critic aux loss, actor trust head) does not survive proper ablation. Trust head treats Monitor as noise. - **DLR sub-hypothesis:** VALIDATED. DLR cross-agent predicates in the critic give a small ($\sim +0.2\%$ relative) but reproducible, non-shrinking signal-specific contribution. n=30 confirms n=5 (same magnitude).

The right framing for the paper is “DLR predicates (not Monitors) in the critic are the right architectural choice for cross-agent signal in cooperative MARL at this compute scale.” Full 6-pathway synthesis: `experiments_log/2026-07-29-y2-final-6-pathway.md`.

H6: Joint Monitor failure is monotonic with PPO updates

Statement: The discrimination power of a joint Monitor decreases monotonically as PPO updates accumulate, due to the policy gradient dragging the Monitor’s signal.

Status: REFUTED (5-seed instrumented, 10K PPO steps)

Evidence (full log: `experiments_log/2026-07-28-h6-instrumented-5seed.md`):

- 5 seeds x 5 evaluation points each (2048, 4096, 6144, 8192, 10240 PPO steps). - Held-out set: 20 rollouts collected ONCE at the start, then reused. - Per-seed Spearman rho between PPO step and heldout Monitor AUROC: - seed 0: -0.894 ($p=0.04$) - VALIDATED - seed 1: +0.894 ($p=0.04$) - REFUTED - seed 2: -0.707 ($p=0.18$) - PARTIAL - seed 3: +0.975 ($p=0.005$) - REFUTED - seed 4: +0.447 ($p=0.45$) - REFUTED - 3/5 seeds show INCREASING AUROC, not decreasing. - Aggregate: mean rho=+0.143, sd=0.887.

Interpretation: The pre-registered mechanism behind H1 (frozen > joint) is NOT ‘joint Monitor loses discrimination over PPO updates’. The actual behaviour is mixed: joint Monitor can increase in AUROC and still fail as a reward signal, because the failure concept it learns is policy-coupled. H1 itself (frozen 0.796 vs joint 0.072 at 100K PPO) is still validated; only H6’s proposed mechanism is refuted.

Implication for Y1 paper: H1 framing stays; the proposed mechanism (‘joint AUROC degrades with PPO’) should be removed or reframed as ‘joint Monitor learns a policy-coupled failure concept that does not transfer as a reward signal even when AUROC is high’.

Y2 follow-up: investigate what failure concept the joint Monitor learns (visualise its predictions vs frozen Monitor on held-out data) and whether early

stopping on the joint Monitor recovers the frozen Monitor’s quality.

H7: Reference Monitor + Evidence Chain (V1 governance)

Statement: A reference Monitor that emits structured events into a hash-chained evidence log achieves >95% tamper-detection rate and zero execution of PEP-denied actions, on a deterministic scripted agent.

Status: [VALIDATED] **VALIDATED** (GovBench H1+H2, 7 seeds)

Evidence (from experiments_log/2026-07-28-govbench): - H1: violation_rate 0.000 (with PEP) vs 1.000 (no PEP), n=7 - H2: tamper_detected 1.000, n=7 - Audit precision: 0.143 with PEP (limitation noted: more lures needed)

Y2 follow-up: combine with DLR predicates for symbolic verification in evidence chain. Test on real LLM backend.

H8: A2A cross-agent trust gate intercepts impersonation

Statement: An executor that consults an external trust registry (verified trust) instead of trusting the requester’s *claimed* trust intercepts adversary A3 (impersonation, claims high trust).

Status: [VALIDATED] **VALIDATED** (GovBench H3, 7 seeds)

Evidence (from GovBench): - H3: intercept_rate 1.000 (with gate) vs 0.000 (no gate), n=7 - executed_violations 0.000 (with gate) vs 1.857 (no gate) - Legitimate delegation success: 1.000 (utility preserved)

Y2 follow-up: integrate with Phase 2 DMC (DMC broadcasts its Monitor signals as trust evidence). Test on real LLM agent + WebArena.

H9: Self-improvement loop with Monitor feedback

Statement: An agent whose policy is **updated** based on Monitor feedback (not just shaped reward) demonstrates measurable improvement in failure prediction AUROC over training, without sacrificing task performance.

Status: [OPEN] **OPEN** (Y3 work – multi-step self-modification)

Evidence so far: - Y1.3 is a 1-step self-modification: Monitor -> reward shaping -> PPO update - After Y1.3 training, Monitor accuracy is not re-evaluated - 2-step self-modification (Monitor re-trained on Y1.3 policy) not tested - Full self-improvement loop (Y1.3 -> new Monitor -> new PPO -> ...) not built

Y2/Y3 follow-up: implement 2-step self-improvement loop; verify Monitor accuracy improves without task regression.

Summary table

H	Status	Key result	Y2 work
H1	[VALIDATED] VALIDATED	frozen > joint, delta=0.724	cross-env
H2	[VALIDATED] VALIDATED	Y1.3 +50, p<0.001	cross-env, longer training
H3	[VALIDATED] VALIDATED	DLR 97.8% 4-env mean	harder envs, learned predicates
H4	[VALIDATED] VALIDATED	Slot-Monitor 0.989 vs 0.796	5-seed validation
H5	partial- REFUTED	5/6 REFUTED; v8 dlr_only (DLR in critic) +0.1447, t=+3.216, p<0.005, 20/30 pos at n=30; Monitor sub-H REFUTED, DLR sub-H VALIDATED	DONE (6-pathway + n=30 conf)
H6	REFUTED	joint AUROC does NOT decrease; 3/5 seeds increase (Spearman rho=+0.14 mean)	remove mechanism from H1 framing
H7	[VALIDATED] VALIDATED	PEP H1 + tamper H2	DLR + real LLM
H8	[VALIDATED] VALIDATED	A2A gate intercepts A3	DMC integration
H9	[OPEN] OPEN	Y1.3 is 1-step	2-step + loop

4/9 hypotheses validated (H1, H2, H3, H4, H7, H8 actually 6). **1/9 partial** (H6). **3/9 open** (H5, H6, H9, H7, H8, H9).

Wait – re-counting: H1 VALID, H2 VALID, H3 VALID, H4 VALID, H7 VALID, H8 VALID = **6 validated**. H6 [triangle] = **1 partial**. H9 = **1 open**. **Total: 6 validated, 2 partial, 1 open.**

Y2 priorities: 1. H5 (DMC) – most impactful, depends on working PPO baseline (have MADDPG) 2. H9 (self-improvement loop) – long-term, depends on Y1.3 3. H6 (instrumented logging) – easy, can do in Y1 cleanup

[End of 9-hypothesis framework. ~6 KB.]

Cross-reference: H1-H10 status table

This document covers H1-H9. H10 (LLM self-monitoring, Y4 paper) was added 2026-07-30 and is tracked in `papers/HYPOTHESIS_STATUS.md`, which supersedes this table by adding H10 and updating H5 with the n=100 follow-up data.

For the Y3 paper (H5 multi-agent) and Y4 paper (H10 LLM self-monitoring) details, see:

- `papers/monitor_signal_vs_dlr_6pathway.md` (Y3)
- `papers/project_g_v0_5_h10_paper.md` (Y4)

Status summary as of 2026-07-31:

- H1, H2, H3, H4, H7, H8: VALIDATED (Y1)
- H1.4, H5, H6, H10: REFUTED (Y1, Y3, Y4)
- H9: OPEN (Y3 follow-up in progress)